



Danshu Jiang

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Chinese, English

BACKGROUND

ABOUT

Undergraduate Biotechnology student at Hunan Agricultural University with hands-on research experience in plant tissue culture, virus detection, and virus-free technology design. Passionate about bridging biology and computation—applying Python and AI tools to solve biological problems. Seeking opportunities in bioinformatics to combine wet-lab expertise with computational analysis.

PROJECTS EXPERIENCE

Virus-Free Technology for Longya Lily (*Lilium brownii*), Hunan Agricultural University

- Present

Designed and optimized a virus-free propagation protocol for Longya Lily, an economically important cultivar. Integrated meristem tip culture with thermotherapy and chemotherapy to eliminate viral pathogens.

- Established shoot tip isolation and meristem culture protocols for *Lilium brownii*
- Applied thermotherapy (37°C, 4 weeks) combined with ribavirin chemotherapy to enhance virus elimination efficiency
- Achieved regeneration of virus-free plantlets with improved vigor compared to infected donor material
- Designed the experimental workflow from explant sterilization through acclimatization

Virus Detection in Lily – LMoV, LSV & CMV, Hunan Agricultural University

- Present

Screened lily samples for three major viral pathogens (Lily Mottle Virus, Lily Symptomless Virus, Cucumber Mosaic Virus) using molecular diagnostic techniques. Validated virus-free status of regenerated plants.

- Performed RT-PCR detection of LMoV, LSV, and CMV from leaf tissue samples
- Extracted total RNA and synthesized cDNA for multiplex screening
- Compared virus incidence rates between field-collected and in-vitro-regenerated material
- Documented detection protocols and maintained experimental records for reproducibility

Plant Tissue Culture & Regeneration Systems, Hunan Agricultural University

- Present

Established and maintained in-vitro culture systems for lily and other horticultural crops. Investigated the effects of plant growth regulators on shoot proliferation and root induction.

- Prepared MS media with varying concentrations of 6-BA, NAA, and IBA for organogenesis
- Monitored and controlled aseptic conditions to minimize contamination rates below 5%
- Optimized subculture intervals and rooting protocols across multiple lily genotypes
- Acclimatized tissue-cultured plantlets in greenhouse conditions with >85% survival rate

TNF Gene Bioinformatics Analysis, Hunan Agricultural University

- Present

Performed computational analysis of Tumor Necrosis Factor (TNF) gene across multiple species. Applied bioinformatics tools for sequence alignment, conserved domain identification, and phylogenetic analysis to understand evolutionary relationships and functional implications.

- Retrieved TNF gene sequences from NCBI GenBank across multiple species
- Performed multiple sequence alignment using Biopython and Clustal Omega
- Identified conserved domains and predicted protein functional sites
- Constructed phylogenetic tree to analyze TNF gene evolutionary relationships across species

AI-Assisted Experimental Design for Lily Virus Elimination, Hunan Agricultural University

- Present

Leveraged large language models (ChatGPT, Claude) to optimize virus elimination protocols for lily. Applied AI tools for literature review, primer design, data analysis, and experimental documentation, significantly accelerating the research workflow.

- Used ChatGPT and Claude Code to screen and summarize relevant virology literature, reducing literature review time by 60%
- Applied LLM-assisted analysis of LMoV/LSV/CMV genome sequences to guide RT-PCR primer design
- Maintained version-controlled experimental protocols and data records on GitHub
- Employed AI tools for statistical analysis, figure generation, and scientific writing

SKILLS

Programming & Bioinformatics

- Python
- Biopython
- Pandas
- Data Analysis
- Sequence Analysis

AI & Productivity Tools

- ChatGPT
- Claude Code
- Large Language Models
- AI-assisted Research
- Prompt Engineering

Wet-Lab Techniques



RT-PCR

RNA/DNA Extraction

Gel Electrophoresis

Tissue Culture

Aseptic Technique

Virus Detection

Collaboration & Version Control

GitHub

Git

Scientific Writing

Protocol Documentation

EDUCATION

Biotechnology, Bachelor of Science, Hunan Agricultural University

Sep, 2022 - Jun, 2026

Molecular Biology

Genetics

Plant Tissue Culture

Virology

Bioinformatics

Biochemistry

Cell Biology

Microbiology

INTERESTS

Bioinformatics

Genomic Data Analysis

Transcriptomics

Viral Genomics

Computational Biology

Plant Biotechnology

Virus-Free Propagation

Molecular Breeding

Germplasm Conservation

